

Troubleshooting Tools

1) Open the command prompt and use the ipconfig tool to find your computer's IPv4 address and default gateway, then take a screenshot of the output.

Step 1: Open Command Prompt

- Press Windows + R, type cmd, and press Enter.

Step 2: Run ipconfig

- Type the following command:
`ipconfig`
- Find the active network adapter and note the IPv4 Address and Default Gateway.

Result

- IPv4 Address: 192.168.75.1
- Default Gateway: 192.168.31.1

2) Use the ping command to check the connectivity between your computer and www.google.com, then note down the average response time shown.

Step 1: Run the Ping Command

- Open Command Prompt and type:

```
ping www.google.com
```

Step 2: Check the Result

- Look at the replies received from Google.
- At the bottom of the output, note the average response time shown by Windows.

Result

- Average response time: 772 ms
- Packets sent: 4
- Packets received: 4
- Packets lost: 0

Explanation

- The ping command checks whether the destination is reachable and measures the round-trip time between the computer and the destination.

3) Run the tracert (or traceroute on Linux/Mac) command to www.flipkart.com and list the number of hops it takes to reach the destination.

Step 1: Run the Command on Windows

- Open Command Prompt and type:

```
tracert www.flipkart.com
```

Step 2: Count the Hops

- Each numbered line in the tracert output represents a hop through a network device or routing point.
- Record the final hop that reaches the destination, if the destination is reached successfully.

Result

- Number of hops to destination: 15
- Destination reached: Yes

4) Use the nslookup tool to find the IP address of www.spotify.com and explain in one line what this tool helps you verify.

Step 1: Run nslookup

- Open Command Prompt and type:
`nslookup www.spotify.com`

Step 2: Record the Result

- Look for the Address or Addresses returned by the DNS lookup.

Result

- IP address returned: _____ 151.101.211.42 _____

Explanation

- nslookup helps verify DNS name resolution by checking which IP address or addresses are returned for a domain name.

5) A friend says they cannot access Zomato's website from their laptop. Suggest a sequence of three troubleshooting commands (choose from: ping, ipconfig, tracert, nslookup) you would run to diagnose the issue, and briefly state what each command would help you check.

Recommended Troubleshooting Sequence

- 1. ipconfig — Check whether the laptop has a valid IPv4 address, subnet mask, default gateway, and DNS configuration.
- 2. ping — Test connectivity to the default gateway first and then to a reachable Internet destination to determine whether basic network connectivity is working.
- 3. nslookup — Check whether the Zomato domain name resolves to an IP address, helping identify a possible DNS resolution problem.

Why This Order?

- ipconfig checks the local network configuration first.
- ping checks basic connectivity after confirming the local configuration.
- nslookup then checks whether DNS can translate the website name into an IP address.

Optional Further Check

- If DNS resolution and basic connectivity appear normal but the website is still unreachable, use tracert to identify where the route toward the destination may be failing.

Conclusion

- This sequence helps separate local IP configuration problems, network connectivity problems, and DNS resolution problems.